

## A COMBINED $Z$ CROSS SECTION FROM CMS OPEN DATA

$Z \rightarrow \mu^+ \mu^-$  and  $Z \rightarrow \tau_h \tau_h$ , and what happens when you put them together

15 September 2026 • CMS Open Data 2016 G+H,  $\sqrt{s} = 13 \text{ TeV}$ ,  $L = 16.4 \text{ fb}^{-1}$

### The result

- $\sigma(pp \rightarrow Z/\gamma^* \rightarrow \ell\ell, 60 < m_{\ell\ell} < 120 \text{ GeV}) = 1940 \pm 35 \text{ pb}$  per lepton flavour
- the two channels agree:  $\chi^2/\text{ndf} = 0.79/1$ ,  $p = 0.37$
- lepton universality:  $R = \sigma(\tau\tau)/\sigma(\mu\mu) = 1.16 \pm 0.18$
- **combining gains 0.13 %** — the luminosity uncertainty is fully correlated and cannot be combined away

### This deck

1. the result, and why it looks like that
2. strategy: two channels, one shared contract
3.  $Z \rightarrow \mu\mu$  and  $Z \rightarrow \tau_h \tau_h$  in two slides each
4. how the combination is actually done

+ backup: references, full breakdown, variations

# The measurement at a glance

**Data.** CMS 2016 Open Data, Run2016G+H, NanoAODv9.  
 $\sqrt{s} = 13 \text{ TeV}$ ,  $L = 16\,393.381 \text{ pb}^{-1} \pm 1.2\%$  (normtag).

**Two independent measurements**, same luminosity, same simulation, same fitter (TRExFitter v1.8.0), same nuisance-parameter names:

- $Z \rightarrow \mu^+ \mu^-$ : 10.4 M events,  $\delta\sigma/\sigma = 1.8\%$
- $Z \rightarrow \tau_h \tau_h$ : 47.6 k events, 81 % of them jet $\rightarrow\tau_h$  fakes,  $\delta\sigma/\sigma = 16\%$

**Combination.** Covariance (BLUE) of the two, with the shared systematics correlated through the Category contract of fitting/CONVENTIONS.md.

**The answer, and the punchline:** the two agree, and combining them gains **0.13 %** – because the luminosity uncertainty is fully correlated and irreducible.

## Result

$$\sigma(pp \rightarrow Z/\gamma^* \rightarrow \ell\ell, 60 < m_{\ell\ell} < 120 \text{ GeV}) = \mathbf{1940 \pm 35 \text{ pb}}$$

$$= 1940 \pm 0.6 (\text{stat}) \pm 23 (\text{syst}) \pm 12 (\text{acc}) \pm 24 (\text{lumi}) \text{ pb}$$

per lepton flavour, assuming lepton universality

$$\text{channel compatibility } \chi^2/\text{ndf} = 0.79/1, p = 0.37$$

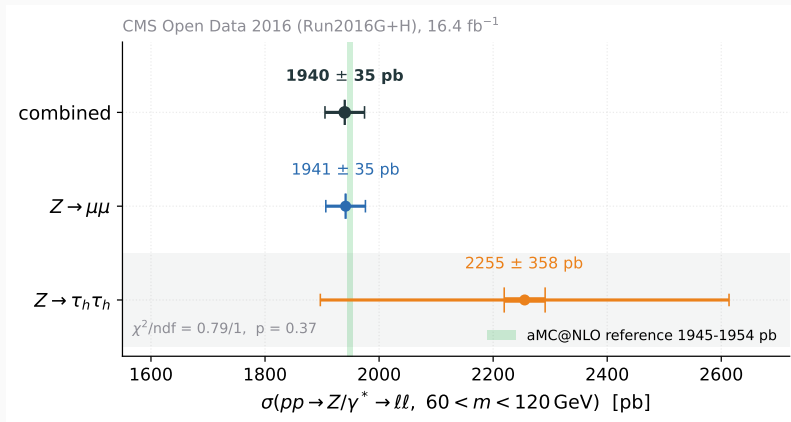
## Lepton universality

$$R = \sigma(Z \rightarrow \tau\tau)/\sigma(Z \rightarrow \mu\mu) = \mathbf{1.16 \pm 0.18}$$

+0.9 $\sigma$  from unity; the luminosity cancels exactly

CMS measures  $1952 \pm 4 \pm 18 \pm 45 \text{ pb}$  for the same quantity (SMP-20-004). We agree within  $0.2\sigma$ .

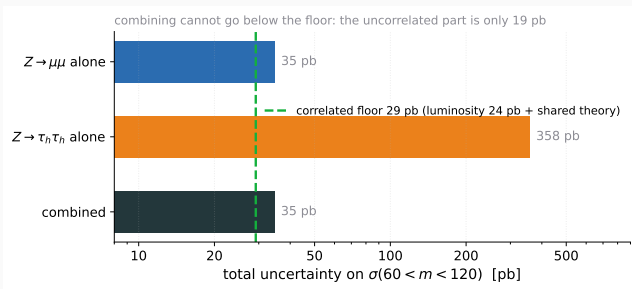
# The combined cross section



Weights:  $\mu\mu + 1.005$ ,  $\tau\tau - 0.005$ . A *negative* weight is normal BLUE behaviour once  $\rho$  exceeds the ratio of the uncertainties ( $0.15 > 0.10$ ): the imprecise measurement becomes a lever on the shared systematic, not an average.

The green band is the aMC@NLO reference, drawn as a *band* because the two channels normalise to 1945 and 1954 pb for the same quantity – a 0.47 % definitional difference we had to handle (backup).

# Why combining gains nothing — and that is the result



$Z \rightarrow \mu\mu$  alone:  $1941 \pm 35$  pb.

Combined:  $1940 \pm 35$  pb.

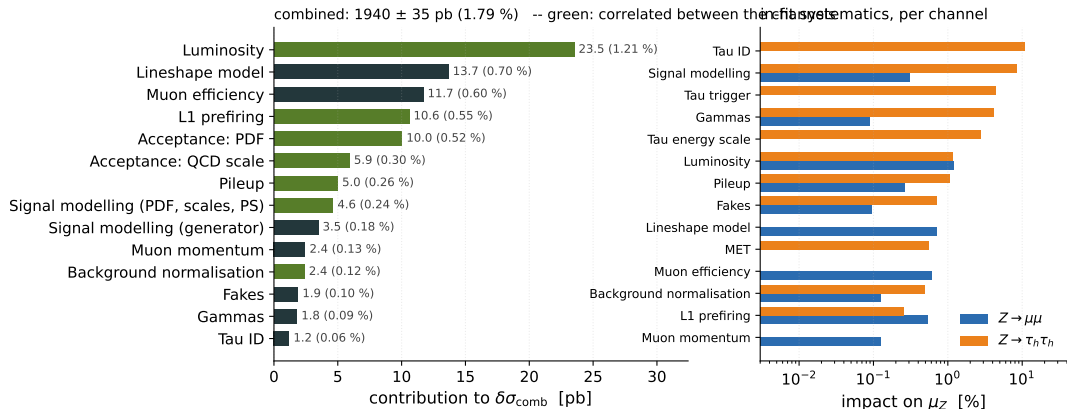
Improvement: 0.13 %.

**Why.** 24 pb of the 35 pb is the luminosity: identical normtag value, identical runs, identical  $\pm 1.2\%$  in both channels. A fully correlated uncertainty cannot be combined away. Add the shared theory terms and the floor is 29 pb.

A third channel of the same precision would not help either. The route to a better  $Z$  cross section from this dataset is a **better luminosity calibration**, not more channels.

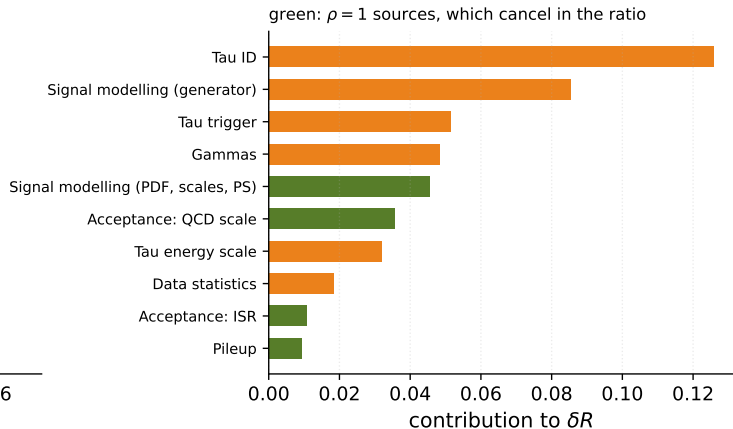
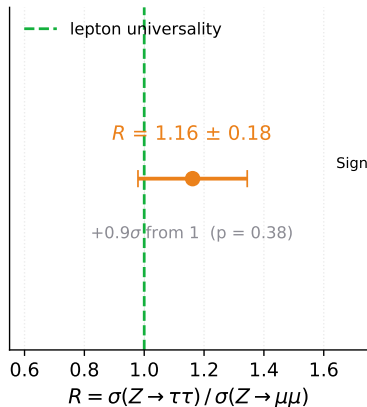
Data statistics contribute  $0.6$  pb –  $0.03\%$ . This measurement was never going to be limited by the number of collisions.

# Where the uncertainty comes from



Left: contribution of each source to the combined uncertainty ( $\delta_s = \sqrt{w^T C_s w}$ , adding in quadrature to the total); green = correlated between the channels. Right: the same systematics as each channel's fit reports them. The  $\tau\tau$  channel's  $\tau_h$  ID (12.5 % on its own  $\mu_Z$ ) survives into the combination as 1.2 pb.

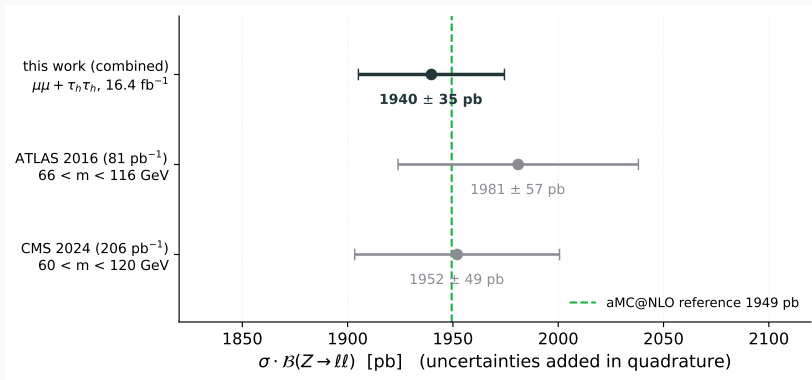
# Lepton universality — what the $\tau\tau$ channel actually buys



The combination *assumes* universality, so it cannot test it. The ratio can — and in the ratio every  $\rho = 1$  source cancels exactly: luminosity, pileup, L1 prefire, and the shared PDF/scale/shower terms.

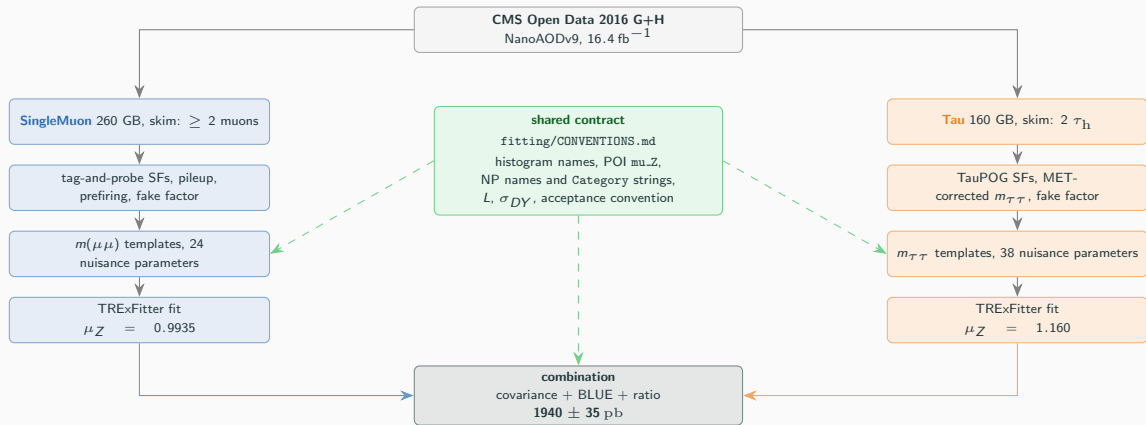
What is left is the  $\tau\tau$  channel's own calibration:  $\tau_h$  ID (12.5 %) and the LO-vs-NLO generator systematic (8.5 %). Improving those improves the *universality test*; it does essentially nothing for the cross section.

## Comparison with published measurements



Agreement with the CMS measurement of exactly the same quantity is within  $0.2\sigma$ ; the ATLAS point uses a narrower window (66–116 GeV) and is shown for scale only. Both published numbers carry a *larger* relative luminosity uncertainty than we do (2.3% and 2.1% against 1.2%) — they are early-Run-2 measurements, while the 2016 G+H norntag luminosity is well measured. Note this is not a fully independent check: the same NNLO reference ( $\sigma_{DY} = 6077.22 \text{ pb}$ ) normalises the simulation behind both of our  $C$  factors.

# How the three pieces fit together



The **green** contract is what makes the combination possible at all: because both channels name 'Lumi', 'Pileup', 'L1Prefiring', 'PDF' ... identically and group them into the same Category, the correlations can be assigned category by category instead of guessed.

# Data and simulation

**Collision data** (Run2016G+H only — 16 of the  $36\text{ fb}^{-1}$  CMS recorded in 2016)

| dataset          | recid         | events |
|------------------|---------------|--------|
| SingleMuon G / H | 30530 / 30563 | 324 M  |
| Tau G / H        | 30532 / 30565 | 156 M  |

Certified with the 2016 golden JSON restricted to runs 278820–284044.

**Luminosity.**  $L = 16\,393.381\text{ pb}^{-1}$ , the *normtag* value from CMS Open Data record 1059. An earlier *brilcalc* call without `--normtag` gave  $16\,290.713\text{ pb}^{-1}$ , 0.63 % low; both channels now use the *normtag* value, which is why the luminosity is *exactly* correlated between them.

**Simulation** (UL16 postVFP, streamed from EOS or read from dCache)

| sample                      | recid             | used for              |
|-----------------------------|-------------------|-----------------------|
| DY M-50 aMC@NLO             | 35669             | signal, both          |
| DY M-50 madgraph            | 35671             | generator syst.       |
| DY M-10to50                 | 35631             | $\tau\tau$ background |
| $t\bar{t}$ 2L2Nu / semilep. | 67801 / 67993     | background            |
| $tW$ top / antitop          | 64895 / 64839     | background            |
| WW, WZ, ZZ                  | 72696/72754/75593 | background            |
| $W$ +jets aMC@NLO           | 69745             | $\tau\tau$ background |

**Normalisation.**  $\sigma(Z/\gamma^* \rightarrow \ell\ell, m > 50\text{ GeV}) = 6077.22\text{ pb}$  (NNLO), divided by 3 per flavour. Both channels'  $\mu_Z$  is a multiplier on *that* number, which is what makes them comparable — and why the residual 0.47 % mismatch matters (backup).

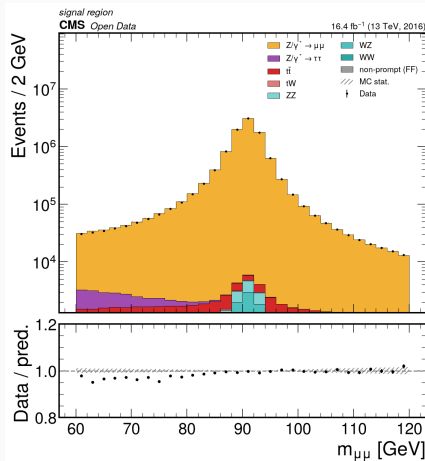
# $Z \rightarrow \mu^+ \mu^-$ : a 1.8 % measurement with 10 million events

**Selection.** HLT\_IsoMu24 || HLT\_IsoTkMu24; exactly two tight, isolated, opposite-sign muons,  $p_T > 26/20$  GeV,  $|\eta| < 2.4$ ,  $60 < m(\mu\mu) < 120$  GeV; FSR photons recovered from the FsrPhoton collection.

**Corrections, all measured in this data.**

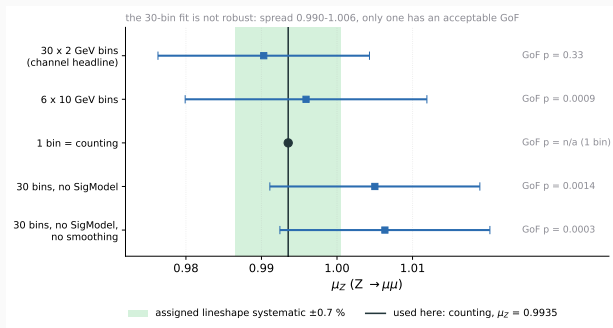
- ID / isolation / trigger scale factors from tag-and-probe fits on the Z peak (20 M pairs), with trigger-object matching
- pileup weights from the Open Data luminosity table, L1-prefiring weights
- Z-peak muon momentum calibration per  $|\eta|$  bin
- non-prompt muons from a same-sign fake factor (3.8 k events, 0.04 % of the yield)

**Backgrounds** are tiny: 69 k of 10.4 M events (0.66 %), dominated by  $t\bar{t}$  (32 k) and  $Z \rightarrow \tau\tau$  (11 k).



The fitted distribution:  $30 \times 2$  GeV bins. Note the ratio panel — data sit 3–4 % below the prediction at 62–78 GeV. That deficit is the reason the next slide exists.

## $Z \rightarrow \mu\mu$ : which number to take — the channel's own review changed it



The channel's headline is a 30-bin shape fit,  $\mu_Z = 0.9903 \pm 0.0140$ . Its own review (`z-mumu/REVIEW.md`, F3) then showed the fit is **not robust**:  $\mu_Z$  moves over 0.990–1.006 with the binning and with whether the SigModel template is in.

Only the nominal configuration has an acceptable goodness of fit, and it gets there by pulling SigModel  $0.5\sigma$  and constraining it to 10 % of its prior — while that template is itself flawed (the powheg sample has a generator cut at  $m_{\text{LHE}} < 120$  GeV, depleting its two highest bins by 10 % and 33 %).

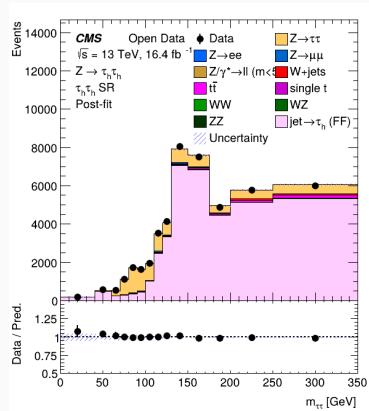
**We follow the review's recommendation:** take the **counting extraction** ( $\sigma_{\text{fid}} = 794.4$  pb,  $\mu_Z = 0.9935$ ) and add  $\pm 0.7\%$  as a lineshape systematic. Using the shape fit instead moves the combination by  $-6.5$  pb — reported as a variation, not used.

# $Z \rightarrow \tau_h \tau_h$ : a 16 % measurement where 81 % of the data is background

**Selection.** Double- $\tau_h$  trigger (different path in G and H), both legs HLT-matched; two opposite-sign  $\tau_h$  with  $p_T > 40$  GeV,  $|\eta| < 2.1$ , decay modes 0/1/10/11, DeepTau2017v2p1 Medium vs jets;  $e/\mu$  veto keeps it orthogonal to the other channels.  $\Rightarrow$  47 586 events, of which  $\sim 81\%$  are jets faking a  $\tau_h$ .

**No QCD simulation is used.** The fake background comes from data: a fake factor  $FF(era, DM, N_{jets}, p_T)$  measured in a same-sign region and applied to opposite-sign events whose leading  $\tau_h$  fails Medium, times an OS/SS correction of 1.07–1.13 from the anti-isolated sideband.

**Mass.** The visible di- $\tau$  mass is not a  $Z$  mass. A per-event likelihood over the  $\tau$  visible-energy fractions using the MET covariance gives  $m_{\tau\tau}$  with scale 0.99 and resolution 11 % (against 0.80 / 13 % for  $m_{vis}$ ).



Post-fit  $m_{\tau\tau}$ , 14 bins, 0–350 GeV. Pink = the data-driven jet  $\rightarrow \tau_h$  fake estimate; orange =  $Z \rightarrow \tau\tau$ . The shape fit is essential: a counting experiment here gives 1812 pb because the high-mass sideband is what fixes the fake normalisation (it pulls it down by 2.8 %, moving  $\sim 1100$  events into the signal).

# $Z \rightarrow \tau_h \tau_h$ : result and what limits it

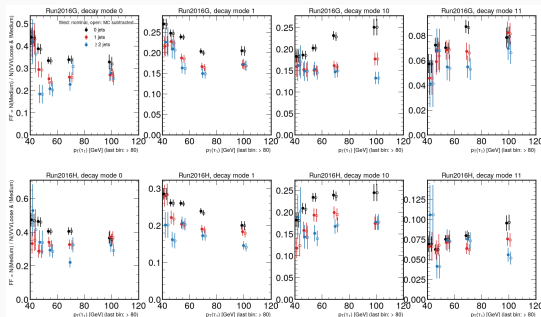
$$\mu_Z = 1.160^{+0.192}_{-0.163}$$

$$\sigma(60-120) = 2255 \pm 36^{+371}_{-315} \pm 83 \text{ pb}$$

goodness of fit  $p = 0.18$

| source                             | impact on $\mu_Z$ |
|------------------------------------|-------------------|
| $\tau_h$ ID                        | 12.5 %            |
| signal modelling (LO vs NLO 8.5 %) | 9.8 %             |
| $\tau_h$ trigger                   | 5.1 %             |
| MC statistics                      | 4.8 %             |
| $\tau_h$ energy scale              | 3.2 %             |
| luminosity                         | 1.3 %             |
| data statistics                    | 1.8 %             |

Acceptance  $A = 0.23 \%$  with a  $3.7 \%$  uncertainty — the double  $40 \text{ GeV}$  cut on the *visible*  $\tau$  momentum sits on the  $Z$   $p_T$  tail, where the QCD scale dependence is  $3.4 \%$ .



The fake factors, binned in era, decay mode,  $N_{\text{jets}}$  and  $p_T$ . Without the  $N_{\text{jets}}$  binning the same-sign closure is off by 10–25 % per jet bin; without the era binning by  $\pm 5 \%$  (the HLT  $\tau$  isolation changed between G and H).

# What is combined, and how

## The observable.

$\sigma(pp \rightarrow Z/\gamma^* \rightarrow \ell\ell, 60 < m_{\ell\ell} < 120 \text{ GeV})$  per lepton flavour, assuming lepton universality:

$$\sigma_c = \hat{\mu}_c \times \sigma_c^{\text{pred}}(60-120)$$

**Not**  $\mu_Z$  itself: the two channels normalise to references 9 pb apart for the same quantity (backup), so averaging the  $\mu$ 's would average two different things.

**The covariance**, source by source,

$$C = \sum_s C_s, \quad (C_s)_{ij} = \rho_s s_i s_j$$

then standard BLUE,

$$w = \frac{C^{-1}u}{u^T C^{-1}u}, \quad \hat{\sigma} = w^T x, \quad \text{Var} = \frac{1}{u^T C^{-1}u}$$

with  $\chi^2$  (ndf=1) for compatibility, and the combined uncertainty decomposed as  $\delta_s = \sqrt{w^T C_s w}$ .

## Two details that are easy to get wrong.

*In-fit systematics scale with the prediction*, because  $\mu_Z$  multiplies a fixed number:  $\delta\sigma = s_{\text{group}} \times \sigma^{\text{pred}}$ . *The acceptance is multiplicative*,  $\delta\sigma = a \times \sigma^{\text{meas}}$ , since  $\sigma = \sigma_{\text{fid}}/A$  — and  $A$  is outside both likelihoods (0.61 % for  $\mu\mu$ , 3.7 % for  $\tau\tau$ ), so the combination has to add it.

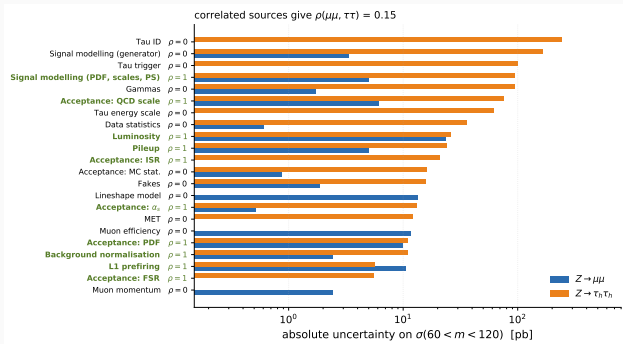
Because the acceptance is multiplicative, the BLUE is *iterated*: the terms are re-evaluated at the combined value until stable (4 iterations,  $< 0.1$  pb).

**Cross-checked** against an explicit profile likelihood with one nuisance per shared source,

$$-2 \ln L = \sum_c \frac{(x_c - \sigma - \sum_s d_{cs} \theta_s)^2}{u_c^2} + \sum_s \theta_s^2$$

which additionally carries the  $\tau\tau$  channel's asymmetric error. Same answer:  $1939^{+35}_{-35}$  against  $1940 \pm 35$  pb.

# The correlation model — the only judgement in the whole thing



$\rho = 1$  — derived once, used by both

- Luminosity: same normtag value, same runs, same  $\pm 1.2\%$
- Pileup, L1 prefire: same profile, same maps, same code
- Background normalisation: literally the same XS\_\* parameters on the same MC samples
- PDF,  $\alpha_s$ , QCD scale, PS: same Hessian members and the same 7-point envelope of the *same* aMC@NLO sample

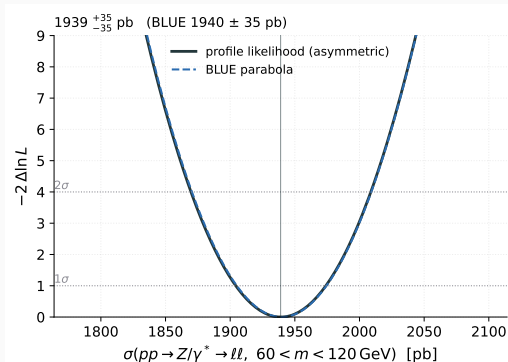
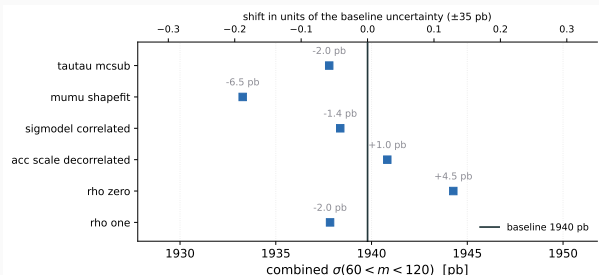
$\rho = 0$  — different objects, methods or events

- muon vs  $\tau_h$  calibrations: no common input
- Gammas: same sample, but  $\mu\mu$  and  $\tau_h \tau_h$  select disjoint events
- fakes, MET, the  $\mu\mu$  lineshape term
- data statistics: *disjoint* primary datasets, orthogonal triggers

**The one genuine split.** Both channels call their generator systematic SigModel, so a MultiFit would correlate them by name — but  $\mu\mu$  compares two NLO generators (0.2% on C) and  $\tau\tau$  compares LO with NLO (7.3%). We split the category by the ranking fraction and decorrelate that part.

$\Rightarrow \rho(\mu\mu, \tau\tau) = 0.15$ .

# Cross-checks: the model is not what limits us



Every variation moves the answer by less than  $0.2\sigma$ . The two rows that *bracket the entire correlation model* — all systematics uncorrelated, all fully correlated — span 6.5 pb, a fifth of the total uncertainty. The largest single variation is not a correlation choice at all: it is which  $\mu\mu$  extraction to take ( $-6.5$  pb).

Right: the profile likelihood and the BLUE parabola lie on top of each other — the honest way of saying that the  $\tau\tau$  channel's asymmetry does not matter here.

# What this is not, and what is left to do

## Not a TRExFitter MultiFit

The channel workspaces are not committed to the repository and were not rebuilt, so the shared nuisance parameters are correlated through their Category totals rather than *profiled jointly*. The fitting is still TRExFitter's — it produced both inputs.

The MultiFit config is generated and syntax-checked (`combination/fit/comb.config`) and one command away once the workspaces exist:

```
git submodule update --init --recursive
bash fitting/build_trexfitter.sh
cd z-mumu && python run_v2.py --from 4
cd ../z-tautau && python run_all.py --from 3
cd ../combination && python fit/run_multifit.py
```

## What the joint fit would add

- shared NPs profiled together:  $\mu\mu$  data constraining Lumi, Pileup, PDF ... for  $\tau\tau$  as well
- $\tau_h$  ID constrained *in situ* if  $e\tau_h/\mu\tau_h$  were added
- per-bin  $\gamma$  parameters instead of one Gammas number
- the  $\tau\tau$  asymmetry exactly

Expected to be small —  $\mu\mu$  carries 99.5 % of the weight and is already profiled — but it is not measured, so the claim is not made.

## Inherited caveats we did not fix

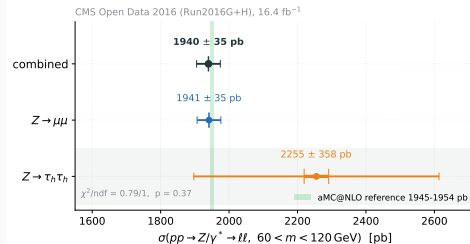
- $\mu\mu$ : MuonReco documented per muon, applied per event (+0.35 % if the former); pileup profile  $\sim 4$  % low
- $\tau\tau$ : the nominal fake factor double-counts genuine  $\tau\tau$  ( $-2$  pb with `mcsb`)
- no  $Z \rightarrow ee$  — out of scope for this pass

# Conclusions

$$\sigma(pp \rightarrow Z/\gamma^* \rightarrow \ell\ell, 60 < m_{\ell\ell} < 120 \text{ GeV}) = 1940 \pm 35 \text{ pb}$$

$$R = \sigma(Z \rightarrow \tau\tau)/\sigma(Z \rightarrow \mu\mu) = 1.16 \pm 0.18$$

1. Two very different analyses of the same  $16.4 \text{ fb}^{-1}$  — 10 M nearly background-free events against 47 k events that are 81 % background — **agree**.
2. The combination is **luminosity-limited**: 24 pb of the 35 pb, fully correlated, irreducible. Combining gains 0.13 %.
3. Data statistics are 0.6 pb. **More collisions would change nothing**; better calibration and more simulated events would.
4. What the  $\tau\tau$  channel really contributes is the **universality test**, where the luminosity cancels exactly.
5. The shared conventions made this possible at all — and the places where they were ambiguous (SigModel, the reference cross section) are exactly where the work went.



All numbers, code and documentation: `combination/result.md`,  
`combination/docs/`. Reproduces from git in seconds with `python`  
`run_combination.py`.

## Backup

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## Backup: the two references differ by 0.47 %

Both channels define  $\mu_Z$  as a multiplier on an aMC@NLO prediction of  $\sigma(pp \rightarrow Z/\gamma^* \rightarrow \ell\ell, 60 < m_{\ell\ell} < 120 \text{ GeV})$  — but they build it differently:

| channel    | construction                                                                | value       |
|------------|-----------------------------------------------------------------------------|-------------|
| $\mu\mu$   | $\sigma_{\text{fid}}^{\text{pred}} / A(60\text{--}120)$<br>799.566/0.409209 | = 1953.9 pb |
| $\tau\tau$ | LHE- $\tau\tau$ subset of the same sample,<br>$60 < m_{\text{LHE}} < 120$   | 1944.9 pb   |

With 72 M generated events the statistical precision on either is 0.02 %, so the 0.47 % is **definitional**, not a fluctuation.

### Why it matters.

- We combine *cross sections*: each  $\hat{\mu}$  is multiplied by its own reference first, so the mismatch shows up as a band on the theory line rather than as a bias.
- A TRExFitter MultiFit with one shared  $\mu_Z$  would *not* handle it — it would fit one parameter against two references.
- Size today: the  $\tau\tau$  weight is 0.5 %, so  $\sim 0.002$  % on the combination. It matters as soon as a second channel of comparable weight appears.

Recorded as a decision in `fitting/CONVENTIONS.md` §6.

## Backup: full uncertainty breakdown of the combined value

| source                           | pb          | of $\sigma$   | corr. |
|----------------------------------|-------------|---------------|-------|
| Luminosity                       | 23.51       | 1.21 %        | yes   |
| Lineshape model ( $\mu\mu$ )     | 13.65       | 0.70 %        | no    |
| Muon efficiency                  | 11.68       | 0.60 %        | no    |
| L1 prefire                       | 10.58       | 0.55 %        | yes   |
| Acceptance: PDF                  | 10.02       | 0.52 %        | yes   |
| Acceptance: QCD scale            | 5.91        | 0.30 %        | yes   |
| Pileup                           | 4.99        | 0.26 %        | yes   |
| Sig. modelling (PDF, scales, PS) | 4.62        | 0.24 %        | yes   |
| Sig. modelling (generator)       | 3.46        | 0.18 %        | no    |
| Muon momentum                    | 2.43        | 0.13 %        | no    |
| Background normalisation         | 2.38        | 0.12 %        | yes   |
| Fakes                            | 1.88        | 0.10 %        | no    |
| Gammas (MC stat.)                | 1.79        | 0.09 %        | no    |
| Tau ID                           | 1.18        | 0.06 %        | no    |
| Acceptance: MC stat.             | 0.87        | 0.05 %        | no    |
| Data statistics                  | 0.63        | 0.03 %        | no    |
| Tau trigger                      | 0.48        | 0.02 %        | no    |
| Acceptance: $\alpha_s$           | 0.47        | 0.02 %        | yes   |
| Tau energy scale                 | 0.30        | 0.02 %        | no    |
| <b>total</b>                     | <b>34.7</b> | <b>1.79 %</b> |       |

Contributions add in quadrature:  $\delta_s = \sqrt{w^T C_s w}$  with  $C_s$  the covariance of that source alone.

Reading the table:

- the top entry and three of the top seven are **correlated**, so they set a floor combining cannot touch;
- the second entry exists only because the  $\mu\mu$  shape fit is not under control — fixing the SigModel template would likely remove most of 13.7 pb;
- everything the  $\tau\tau$  channel contributes sums to 1.2 pb;
- the data statistical uncertainty is 0.6 pb, four orders of magnitude below the cross section.

Generated table: `combination/result.md`, machine-readable version `combination/output/combination.result.json`.

## Backup: variations in numbers

| variation               | $\sigma$ [pb] | shift | what it changes                                    |
|-------------------------|---------------|-------|----------------------------------------------------|
| <b>baseline</b>         | <b>1939.8</b> | —     | $\mu\mu$ counting, $\tau\tau$ nominal, model of §4 |
| $\tau\tau$ mcsb         | 1937.8        | −2.0  | fake factor with genuine- $\tau$ subtraction       |
| $\mu\mu$ shape fit      | 1933.3        | −6.5  | the channel's 30-bin fit instead of counting       |
| SigModel correlated     | 1938.4        | −1.4  | generator comparison shared between channels       |
| acc. scale decorrelated | 1940.8        | +1.0  | acceptance QCD-scale term independent              |
| $\rho = 0$ everywhere   | 1944.3        | +4.5  | all systematics uncorrelated (naive)               |
| $\rho = 1$ everywhere   | 1937.8        | −2.0  | all systematics fully correlated                   |

The  $\rho = 0$  and  $\rho = 1$  rows bracket every possible correlation assignment: they span 6.5 pb,  $0.19\sigma$ . Whatever one believes about the correlations, the answer is the same to within a fifth of its uncertainty. Note also that  $\rho = 1$  and the  $\tau\tau$  mcsb variant happen to land on the same value by coincidence.

## Backup: the channels' own numbers

|                          | $Z \rightarrow \mu^+ \mu^-$     | $Z \rightarrow \tau_h \tau_h$       |
|--------------------------|---------------------------------|-------------------------------------|
| selected events          | 10 378 567                      | 47 586                              |
| background               | 68 821 (0.66 %)                 | 40 248 (85 %, prefit)               |
| A (fiducial / 60–120)    | 0.4092                          | 0.002314                            |
| C (reco / fiducial)      | 0.7917                          | 0.1068                              |
| fit variable             | $m(\mu\mu)$ , $30 \times 2$ GeV | $m_{\tau\tau}$ , 14 bins, 0–350 GeV |
| nuisance parameters      | 24 (+30 $\gamma$ )              | 38 (+14 $\gamma$ )                  |
| $\mu_Z$                  | 0.9935 (counting)               | $1.160^{+0.192}_{-0.163}$           |
| $\sigma_{\text{fid}}$    | 794.4 pb (pred. 799.6)          | 5.22 pb (pred. 4.50)                |
| $\sigma(60\text{--}120)$ | $1941 \pm 35$ pb                | $2255 \pm 358$ pb                   |
| goodness of fit          | — (1 bin)                       | $p = 0.18$                          |

The asymmetry is the point: one channel is a precision measurement whose entire difficulty is calibration, the other is a needle-in-a-haystack whose entire difficulty is the background. They share a luminosity, a signal sample and a fitter, and nothing else.

## Backup: where everything lives

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|                                |                                                            |
|--------------------------------|------------------------------------------------------------|
| combination/result.md          | the result, generated from the JSON — never edited by hand |
| combination/docs/00--04        | overview, inputs, correlation model, method, results       |
| combination/comb/              | inputs, model, blue, likelihood, ratio, plots, report      |
| combination/run_combination.py | orchestrator; --check runs 6 closure tests                 |
| combination/fit/comb.config    | the TRExFitter MultiFit, generated and validated, not run  |
| combination/output/            | combination_result.json, plots/*.pdf                       |

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|                        |                                                                   |
|------------------------|-------------------------------------------------------------------|
| z-mumu/, z-tautau/     | the channel analyses, each with its own README, docs/, handoff.md |
| z-mumu/REVIEW.md       | the review that changed which $\mu\mu$ number we use              |
| fitting/CONVENTIONS.md | the shared contract; §6 records what the combination decided      |
| presentation/          | this deck (bash build.sh)                                         |

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Closure tests behind --check: each published channel number is re-asserted; a single-channel BLUE reproduces that channel exactly; two identical measurements with  $\rho = 0$  give error/ $\sqrt{2}$  and equal weights; with every systematic correlated only the statistical part averages; and the profile likelihood reproduces BLUE to  $10^{-3}$  pb.